Sensitivity Analysis Protocol

Validation Methodology for the Kuramoto Phase Engine in NEUROCARDIA

Companion document to the NEUROCARDIA White Paper

Version 0.2 — Protocol and Results

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Revision history. This is revision 0.2 of the *Sensitivity Analysis Protocol*. Revision 0.1 (October 14, 2026) specified the protocol, the five experimental arms, and the acceptance criteria, but reserved the results section for a subsequent revision. Revision 0.2 populates the results section, adds the residual AUC as a fourth evaluation metric, introduces a companion acceptance criterion for the residual diagnostic, and reports the finding that trajectory robustness and residual robustness are disjoint. The protocol specification of revision 0.1 is unchanged.

Abstract

The NeuroCardia diagnostic engine applies a Kuramoto network of $N = 16$ coupled oscillators to the RR tachogram of a single-lead ECG recording, producing a Cardiac Synchronization Index (CSI, $r(t)$) and a family of early warning signals (EWS). Three parameter choices distinguish this engine from the standard Kuramoto literature: a deterministic frequency distribution drawn from the non-trivial Riemann zeros, a carrier frequency five orders of magnitude above the cardiac band, and logarithmic rather than linear spacing between adjacent frequencies. This document specifies the formal sensitivity analysis protocol that determines whether the resulting CSI and EWS are robust against these choices, and reports the results of running the protocol against the MIT-BIH Arrhythmia Database and a paired corpus of normal sinus and atrial fibrillation recordings.

The principal findings are two. First, the absolute value of the CSI is not robust to the frequency distribution at any coupling strength where the metric has dynamic range. The mean Pearson correlation between the CSI trajectory under the Riemann-zero configuration and under any alternative configuration is at most 0.17 at $K = 2$, and reaches the pre-registered threshold of 0.80 only for the linear-spacing distribution at $K = 30$, where the network is saturated. Second, the residual diagnostic that the companion papers report is not robust to the frequency distribution either: three of five distributions exhibit the residual at $K = 2$ (AUC 0.84–0.89), and two do not (AUC 0.54–0.56). The two robustness properties are disjoint: no monotone relation exists between trajectory correlation and residual AUC across the five arms.

The protocol's purpose is both scientific defensibility and satisfaction of the analytical validation requirements of the IEC 62304 / FDA 510(k) pathway.

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1 Introduction

1.1 Context

The NeuroCardia diagnostic engine applies a Kuramoto model of coupled phase oscillators to the RR interval series of a single-lead ECG recording. The model treats the sinoatrial node as a population of coupled pacemaker cells and produces a time-varying order parameter $r(t)$, called the Cardiac Synchronization Index (CSI). A family of derived metrics — variance, lag-1 autocorrelation, skewness, kurtosis — is computed over sliding windows and reported as early warning signals (EWS) of impending rhythm transitions.

The engine is a specific parameterization of the standard Kuramoto model. Three of its parameter choices are inherited from the Convergent reference framework and are not found in the standard literature. This document specifies the protocol by which those choices are tested for robustness, and reports the results.

1.2 Why sensitivity analysis

A diagnostic metric is only as trustworthy as its robustness to arbitrary choices. If the CSI is driven by the physical dynamics of the cardiac rhythm, it should be stable across reasonable variations of the model's free parameters. If it is driven by the specific values of those parameters, then any diagnostic claim derived from it is conditioned on those values, and the model's scientific value is correspondingly reduced.

For a research tool, this is an interesting question. For a medical device, it is a regulatory requirement. The FDA's guidance on analytical validation for SaMD requires that the manufacturer demonstrate that the algorithm computes what it claims to compute, and that its outputs are not artifacts of parameter selection.

1.3 Scope

This document specifies: the three parameter choices under test (Section 2), the experimental design comprising five arms (Section 3), the evaluation metrics applied to each arm (Section 4), the acceptance and falsification criteria (Section 5), the reporting protocol (Section 6), and the results of running the protocol (Section 7).

2 Parameters Under Test

2.1 Deterministic frequency distribution

Standard Kuramoto implementations assign each oscillator a natural frequency drawn from a probability distribution $g(\omega)$ — commonly Gaussian, uniform, or Lorentzian. NeuroCardia instead assigns frequencies deterministically from the non-trivial zeros of the Riemann zeta function. The first 16 such zeros are used for the production network of $N = 16$ oscillators, and the first 24 are tabulated for extended networks.

The author does not claim that the Riemann zeros have a cardiac physiological correlate. The choice reflects a mathematical rather than a physiological motivation. Whether this matters for the diagnostic output is the question this protocol addresses.

2.2 Carrier frequency scale

The reference parameterization places the carrier frequency at $\Omega_T = 587{,}000$ rad/s, five orders of magnitude above the cardiac rhythm (approximately 1 Hz). The dimensionless order parameter $r(t)$ is, by construction, invariant to the absolute scale of Ω_T, because the RR-to-phase mapping and the R-peak phase resets couple the network to relative beat timing only. The normalized frequencies used in production are obtained by

subtracting the carrier and scaling to unit variance; this linear transformation does not change the shape of the distribution.

2.3 Logarithmic frequency spacing

The Riemann zeros grow denser as γ increases. Their spacing is governed by the Riemann-von Mangoldt formula, giving an approximately logarithmic density. This contrasts with the linear spacing of a uniformly distributed frequency set, and with the unimodal clustering of a Gaussian set. Whether the CSI depends on the shape of the frequency distribution, or only on its overall variance, is an empirical question.

3 Experimental Design

3.1 Benchmark corpus

The analysis runs on a curated subset of the PhysioNet reference databases chosen to span the clinically relevant range of cardiac dynamics. For the trajectory sensitivity analysis, 48 records from the MIT-BIH Arrhythmia Database [1] were used. For the residual sensitivity analysis, a paired corpus of nine records from the MIT-BIH Normal Sinus Rhythm Database [2] and nine records from the MIT-BIH Atrial Fibrillation Database [3] was used, each trimmed to the first ten minutes.

All records are annotated. R-peak detection is performed by the Pan-Tompkins module in NeuroCardia [4], then verified against the reference annotations before any CSI computation. Records with $Se < 99\%$ or $PPV < 99\%$ are excluded from the analysis and reported separately.

3.2 Experimental arms

Five arms share an identical preprocessing pipeline (Pan-Tompkins R-peak detection, RR-to-phase mapping) and differ only in the Kuramoto engine frequency distribution.

Arm	Distribution	Notes
A	Riemann zeros	Reference configuration
B	Uniform random	Stochastic; seed 42
C	Gaussian	Deterministic (fixed seed)
D	Physiological scale	Bounded, uniform on a physiological range
E	Linear spacing	Uniform on an integer grid

Table 1: The five experimental arms. All arms use $N = 16$, all-to-all coupling, and the same RR-to-phase mapping.

3.3 Controlled variables

The number of oscillators ($N = 16$), the coupling topology (all-to-all), the integration scheme (Euler with $\Delta t = 0.05$ s), the phase-reset rule (reset oscillator 0 at each R-peak to the cumulative phase), and the R-peak detector (Pan-Tompkins) are held fixed across arms.

4 Evaluation Metrics

4.1 Trajectory correlation

The primary metric is the Pearson correlation between the CSI trajectory $r_A(t)$ under Arm A and the trajectory $r_X(t)$ under each alternative arm X. Because the trajectories may have different lengths if the reset schedule differs, we sample both at the same beat indices and compute the correlation over the shorter of the two.

The secondary trajectory metric is the normalized dynamic time warping (DTW) distance between the two trajectories. Lower values indicate greater similarity.

4.2 Residual AUC

The trajectory correlation measures whether the CSI is well-defined under parameter perturbation. It does not measure whether the CSI is diagnostic. The companion papers [9] identify a specific diagnostic — the residual of the CSI after regressing out the coefficient of variation of the RR series — and report that it separates sinus from AF at $K = 2$ with AUC 0.89 under Arm A.

We therefore add a second criterion: the residual AUC must be ≥ 0.80 for the arm to be considered diagnostically usable. This criterion is independent of the trajectory criterion, and Section 7.4 shows that the two are disjoint.

4.3 Bifurcation and transition timing

Where the downstream diagnostic output is a discrete classification over a sliding window, the concordance between Arm A and each alternative arm is measured by Cohen's Kappa. This metric was specified in revision 0.1 as a secondary criterion. It is reported where applicable; the pilot corpus does not have the annotated transitions required for a full analysis.

5 Acceptance and Falsification Criteria

5.1 Trajectory acceptance

The engine is deemed robust to a parameter if, across all records in the benchmark corpus, the trajectory correlation with Arm A is at least $r = 0.80$, and the normalized DTW distance is at most the within-Arm-A variability measured on repeated runs of the same record.

5.2 Residual acceptance

The engine is deemed diagnostically robust to a parameter if, on the paired sinus/AF corpus, the residual AUC for the corresponding arm is at least 0.80 at the optimum K.

5.3 Partial robustness

If the trajectory correlation falls between $r = 0.60$ and $r = 0.80$, or the residual AUC falls between 0.65 and 0.80, the parameter is deemed partially robust. In this case the white paper is updated to qualify the corresponding claim, and the parameter is flagged as a modeling assumption rather than a physical invariant.

5.4 Falsification

If the trajectory correlation falls below $r = 0.60$, or the residual AUC falls below 0.65, the parameter is deemed not robust for the corresponding criterion. In this case: the white paper's corresponding claim is restricted; the parameterization is designated a model-dependent projection; and the diagnostic value of the metric is reported as conditioned on the specific parameterization.

5.5 Interpretation

The protocol is designed to be falsifiable. A result that confirms robustness is a positive scientific finding. A result that refutes robustness is not a failure of the project — it is an honest characterization of what the metric measures and what it does not. The white paper's regulatory claims are conditional on this outcome.

6 Reporting Protocol

6.1 Pre-registration

The protocol specified in this document is fixed before any sensitivity analysis is run. Subsequent modifications are logged in a revision history, dated, and accompanied by a stated rationale. The purpose is to prevent post-hoc adjustment of the acceptance criteria in response to observed results.

6.2 Outputs

For each record, the analysis produces: a CSV file with one row per arm per seed per metric; a per-record CSV for the residual with one row per arm per record; and a summary table aggregating across the corpus. All outputs are committed to the repository under `validation/reports/` and are reproducible with a single Gradle command.

6.3 Publication

The results of the sensitivity analysis are reported in Section 7 of this document. Where the results are of general scientific interest they are additionally reported in the companion documents [9, 8].

7 Results

7.1 Trajectory robustness

Table 2 reports the mean $\pm$ SD of the Pearson correlation between the CSI trajectory under Arm A and under each alternative arm, at five values of the coupling strength K, across 48 mitdb records.

Three observations.

First, at $K = 2$ — the coupling strength at which the residual diagnostic is strongest — all four alternative arms fail the trajectory criterion, with mean correlations of 0.05–0.17 and standard deviations small relative to the means. Every individual record fails the threshold, not just the mean.

Second, the trajectory correlation rises monotonically with K up to $K = 30$, at which point Arm E meets the 0.80 threshold. But $K = 30$ is the coupling strength at which the CSI saturates: the mean order parameter across the corpus is ≈ 0.998 with variance $\approx 10^{-5}$, and the metric is a monotone function of RR variability. Meeting the trajectory criterion at a saturated coupling strength is not a robustness success.

K	Arm B	Arm C	Arm D	Arm E
2.0	0.052 ± 0.030	0.076 ± 0.042	0.051 ± 0.035	0.168 ± 0.089
5.0	0.250 ± 0.045	0.230 ± 0.031	0.250 ± 0.045	0.611 ± 0.177
10.0	0.386 ± 0.102	0.366 ± 0.085	0.386 ± 0.102	0.778 ± 0.171
30.0	0.458 ± 0.165	0.428 ± 0.152	0.458 ± 0.165	$\mathbf{0.869 \pm 0.131}$
100.0	-0.000 ± 0.005	0.000 ± 0.006	0.001 ± 0.005	0.001 ± 0.005

Table 2: Mean $\pm$ SD Pearson correlation between the CSI trajectory under Arm A and under each alternative arm, across 48 mitdb records. The pre-registered threshold is $r = 0.80$. Only Arm E at $K = 30$ meets it, and that is the coupling strength at which the network is saturated.

Third, at $K = 100$ the trajectory correlation collapses to zero and the DTW distance is the largest in the table. This is the regime in which the Euler integration aliases the sine term: the phase advances by more than π per step and the sine evaluation is effectively random.

The conclusion is that the CSI's absolute trajectory is not robust to the frequency distribution at any coupling strength where the metric has dynamic range.

7.2 Residual robustness

Table 3 reports the residual AUC at $K = 2$ for each of the five arms, on the paired sinus/AF corpus of nine and nine records.

Arm	Distribution	Trajectory ρ vs A	Residual AUC	Residual p
A	Riemann zeros	1.000 (ref)	**0.889**	0.002
B	Uniform random	0.052	0.543	0.986
C	Gaussian	0.076	**0.852**	0.007
D	Physiological scale	0.051	0.556	0.982
E	Linear spacing	0.168	**0.840**	0.010

Table 3: Trajectory correlation and residual AUC at $K = 2$ for each arm. Arms A, C, and E exhibit a diagnostic residual; arms B and D do not. The trajectory correlation and residual AUC are not monotonically related across arms.

Three of five arms exhibit the residual; two do not. The two distributions that fail (uniform random and physiological scale) are both bounded-support distributions, but the linear-spacing arm is also bounded-support and does exhibit the residual, so bounded support is not the distinguishing feature. We do not offer a mechanistic explanation for the split.

The practical consequence is that the frequency distribution is part of the metric's definition, and its choice matters for both the trajectory and the residual.

7.3 Orthogonality of the residual to standard HRV

To test whether the residual carries information beyond standard HRV, we regress it on the two features it is most correlated with in the pilot corpus — the Poincaré SD1/SD2 ratio and the coefficient of variation of the RR series — and compute the AUC of the orthogonalized residual.

The orthogonalized residual retains AUC 0.716. Approximately two thirds of the residual's diagnostic value is attributable to its linear relationship with the two HRV features; one third is not. At $n = 18$ the

Quantity	ρ vs label	AUC
Raw residual	$+0.675$	0.889
Residual $\perp$ $(\mathrm{sd1}/\mathrm{sd2}, \mathrm{cv})$	$+0.375$	0.716

Table 4: Orthogonality of the residual to standard HRV features. The residual retains an AUC of 0.716 after orthogonalization, down from 0.889.

confidence interval on the orthogonalized AUC is wide, and the difference between 0.889 and 0.716 is not statistically significant at this sample size.

7.4 Window-length dependence of the residual

The residual criterion of Section 4.2 was applied to full-record analyses. A subsequent window-length sweep shows that the residual is a record-level quantity: at 30-second windows the AUC is 0.50; at 1–5 minute windows it plateaus at 0.66–0.68; at the full record it is 0.83–0.89. The residual's within-record variance is 1.2–2.5 times its between-record variance, and its window-to-window variation does not correlate with window-level P-wave features. The criterion therefore validates a record-level diagnostic, not a within-record state tracker. The scope of the acceptance criterion is noted here for clarity; it does not change the per-arm results reported below.

7.5 Disjointness of trajectory and residual robustness

Table 5 reproduces the key columns of Tables 2 and 3 side by side.

Arm	Trajectory ρ vs A (K = 2)	Residual AUC (K = 2)
A	1.000	0.889
B	0.052	0.543
C	0.076	0.852
D	0.051	0.556
E	0.168	0.840

Table 5: Trajectory correlation and residual AUC are disjoint. Arm C has the second-lowest trajectory correlation and the second-highest residual AUC. Arm E has the highest trajectory correlation among the alternative arms and the second-highest residual AUC. Arms B and D fail on both metrics. No monotone relation exists between the two.

The pre-registered trajectory criterion, applied to the useful coupling regime, would report a failure for every arm. The residual criterion reports a failure for two of five. The two criteria do not identify the same arms as robust, and neither criterion subsumes the other.

8 Discussion

8.1 What the protocol establishes

The protocol establishes two facts. First, the absolute value of the CSI is not robust to the frequency distribution at any useful coupling strength. This is a negative result about the metric's definition. Second, the

residual diagnostic that the companion papers report is also not distribution-agnostic: three of five tested distributions exhibit it.

Both facts condition the metric's interpretation. The CSI is a parameterized quantity; its value refers to a specific parameterization.

8.2 What the protocol does not establish

The protocol does not establish which distributions are physiologically appropriate, nor whether any of the tested distributions has a claim to physical significance. It does not establish the residual's behavior under nonlinear regression, under alternative orthogonality procedures, or on the full reference corpora. These are subjects for further work.

8.3 Implications for the regulatory pathway

For the FDA 510(k) pathway, the protocol demonstrates that the algorithm's absolute output depends on parameter choices in a documented way, and that the parameter dependence is quantified. The regulatory strategy described in the white paper treats the CSI as a research feature whose interpretation is conditioned on the parameterization, and the residual as a separately validated discriminative feature. Both postures are consistent with the results reported here.

9 Conclusion

This document specifies and reports a formal sensitivity analysis of the NeuroCardia Kuramoto phase engine against the PhysioNet reference databases. The absolute value of the CSI is not robust to the frequency distribution at any coupling strength where the metric has dynamic range. The residual diagnostic at $K = 2$ is exhibited by three of five tested frequency distributions. Trajectory robustness and residual robustness are disjoint properties of the metric, and the protocol's pre-registered trajectory criterion is not sufficient to validate the diagnostic.

The results support the white paper's regulatory posture: the CSI is a research metric whose value is parameterized; the residual is a candidate diagnostic whose robustness is narrower than initially reported but is quantifiable and reproducible.

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